

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and built **production-grade agentic AI systems** using **Python** with **LangGraph** stateful orchestration, implementing multi-step agent workflows for mission-critical operations in regulated financial environments, achieving **99.8% uptime** and reducing manual intervention by **60%**.
- Architected **context management strategies** across long-running agent executions, implementing state persistence, retry logic, and idempotency patterns to handle distributed workflows processing **10K+ daily transactions** with zero data loss.
- Built and maintained **RAG pipelines** leveraging **structured, semi-structured, and unstructured data sources** across **PostgreSQL, Oracle, MySQL, and vector stores**, optimizing retrieval strategies and embedding models to achieve **94% retrieval accuracy** for enterprise knowledge bases.
- Developed **MCP-style tool integrations** and context-aware agent frameworks handling **workflow orchestration concepts** including state management, conditional branching, and error recovery, powering agents that reduced query resolution time by **70%** across cross-functional workflows.
- Deployed and operated **agentic systems in highly regulated environments**, implementing compliance-grade audit trails, data governance, and role-based access controls aligned with financial-services regulatory requirements (GDPR, payments processing standards), supporting **SLA-critical enterprise workflows**.
- Collaborated with enterprise stakeholders to design agent workflows for mission-critical business processes, translating requirements into **scalable Python implementations** using **NoSQL and relational databases**, reducing operational bottlenecks by **45%** and enabling real-time decision support across interconnected business systems.

SKILLS

AI & Agentic Systems – LangGraph, agentic AI systems, stateful orchestration, agent workflows, context management, MCP-style tools, workflow orchestration.

Python & Enterprise Development – Python, FastAPI, Flask, commercial enterprise Python, production-grade systems, data pipelines.

RAG & Data Architecture – RAG pipelines, embeddings, retrieval strategies, vector stores, structured data sources, semi-structured data sources, unstructured data sources, context-aware retrieval.

Databases & Data Systems – PostgreSQL, Oracle, MySQL, NoSQL databases, relational databases, vector databases, data persistence, state management.

Enterprise & Cloud Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD, application monitoring, production support.

Data Science & Analytics – Python data stack (pandas, NumPy, scikit-learn), SQL, data modelling, Power BI, Tableau, statistical analysis.

Domain Expertise – Highly regulated environments, financial-services compliance (GDPR, ISO 20022, SEPA, payments processing), enterprise business systems, change-control management, SLA management.

EDUCATION AND TRAINING

PROJECTS

Multi-Agent Financial Operations Platform with LangGraph Orchestration (HCL Technologies)

- Engineered a production-grade **agentic AI system** using **LangGraph** to orchestrate multi-step financial workflows including transaction validation, compliance screening, and settlement reconciliation, implementing stateful context management across **15+ parallel agent nodes** handling **50K+ daily payment operations**.
- Built **RAG pipeline** integrating **structured transaction databases (PostgreSQL, Oracle)**, **semi-structured regulatory policy documents**, and **vector embeddings** with custom retrieval strategies, enabling agents to resolve compliance queries with **96% accuracy** and reducing manual review overhead by **75%**.
- Implemented **workflow orchestration patterns** for state management, retry logic, and idempotency across long-running agent executions processing **multi-day settlement cycles**, achieving **zero data loss** and **99.8% execution success rate** in a highly regulated payments environment.
- Developed **MCP-style tool integrations** enabling agents to query external banking systems, execute ISO 20022 message flows, and trigger ERP updates, creating a seamless decision-support layer that reduced operational incident response time from **4 hours to 18 minutes**.
- Deployed system on **AWS (Lambda, EC2, CloudWatch)** with **Docker containerization**, automated monitoring dashboards, and compliance audit trails, enabling enterprise stakeholders to safely delegate mission-critical workflows to agentic automation while maintaining regulatory oversight.

SWIFT Payments Reconciliation & Context Management Engine (HCL Technologies)

- Architected context persistence layer for agent-driven **SWIFT MT/MX message reconciliation**, maintaining state across multi-day settlement workflows and implementing conflict-resolution logic that improved reconciliation accuracy by **35%** and eliminated **240+ manual workarounds annually**.
- Integrated **relational (PostgreSQL) and NoSQL databases** with vector embeddings for semantic search across payment metadata, enabling agents to retrieve cross-border payment context, sanctions-screening results, and historical transaction patterns in **<100ms latency** to support real-time agent decision-making.
- Designed **RAG-powered compliance knowledge base** linking PSD2 regulations, SEPA payment schemes, and ISO 20022 standards, reducing agent hallucination and enabling safe automation of compliance-sensitive workflows in a **GDPR-regulated financial services environment**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)